

LATIAN WEEK 4

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1.

Code

```
package main

import "fmt"

func factorial(n int, hasil *int) {
    *hasil = 1
    for i := 1; i <= n; i++ {
        *hasil *= i
    }
}

func permutation(n int, r int, hasil *int) {
    var nFact, nmrFact int
    factorial(n, &nFact)
    factorial(n-r, &nmrFact)
    *hasil = nFact / nmrFact
}

func combination(n int, r int, hasil *int) {
```

```
var nFact, rFact, nmrFact int
factorial(n, &nFact)
factorial(r, &rFact)
factorial(n-r, &nmrFact)
*hasil = nFact / (rFact * nmrFact)
}

func main() {
    var a, b, c, d int
    var p1, c1, p2, c2 int

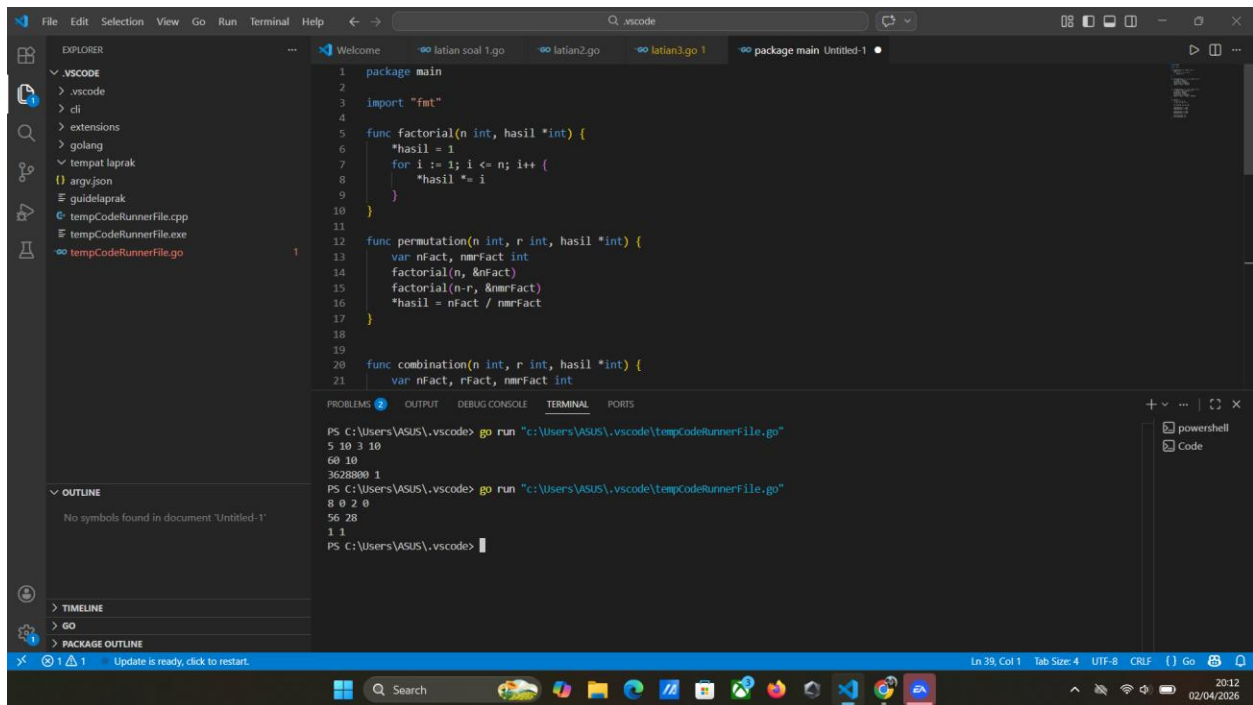
    fmt.Scan(&a, &b, &c, &d)

    permutation(a, c, &p1)
    combination(a, c, &c1)

    permutation(b, d, &p2)
    combination(b, d, &c2)

    fmt.Println(p1, c1)
    fmt.Println(p2, c2)
}
```

Screenshot code



2.

Code

```
package main
```

```
import "fmt"
```

```
func hitungSkor(soal *int, skor *int) {
```

```
    var waktu int
```

```
    *soal = 0
```

```
    *skor = 0
```

```
    for i := 0; i < 8; i++ {
```

```
        fmt.Scan(&waktu)
```

```
        if waktu <= 300 { // 300 menit = 5 jam
```

```
            *soal++
```

```
            *skor += waktu
```

```
    }  
    }  
}  
  
func main() {  
    var nama, pemenangNama string  
    var soalSelesai, totalSkor int  
    var maxSoal, minSkor int  
  
    maxSoal = -1  
    minSkor = 999999  
  
    for {  
        fmt.Scan(&nama)  
        if nama == "Selesai" {  
            break  
        }  
  
        hitungSkor(&soalSelesai, &totalSkor)  
  
        if soalSelesai > maxSoal {  
            maxSoal = soalSelesai  
            minSkor = totalSkor  
            pemenangNama = nama  
        } else if soalSelesai == maxSoal {  
            if totalSkor < minSkor {
```

```

        minSkor = totalSkor

        pemenangNama = nama

    }

}

}

fmt.Printf("%s %d %d\n", pemenangNama, maxSoal, minSkor)

}

```

Screenshot Code

The screenshot shows a VS Code editor with a Go file named `latian4.2.go`. The code defines a `hitungSkor` function that calculates the total score for a given number of questions (`soal`) and a time limit (`waktu`). The `main` function reads input for a name, number of questions, and time limit, then calls `hitungSkor` and prints the results.

```

1 package main
2
3 import "fmt"
4
5 func hitungSkor(soal *int, skor *int) {
6     var waktu int
7     *soal = 0
8     *skor = 0
9     for i := 0; i < 8; i++ {
10        fmt.Scan(&waktu)
11        if waktu <= 300 { // 300 menit = 5 jam
12            *soal++
13            *skor += waktu
14        }
15    }
16 }
17
18 func main() {
19     var nama, pemenangNama string
20     var soalSelesai, totalSkor int

```

The terminal output shows the execution of the program with the following input and output:

```

PS C:\Users\ASUS\.vscode> go run "c:\codingsemester2\latian4.2.go"
Astuti 20 50 301 301 61 71 75 10
Bertha 25 47 301 26 50 60 65 21
Selesai
Bertha 7 294
PS C:\Users\ASUS\.vscode>

```

3.

Code

```
package main
```

```
import "fmt"

func cetakDeret(n int) {
    for {
        fmt.Printf("%d ", n)

        if n == 1 {
            break
        }

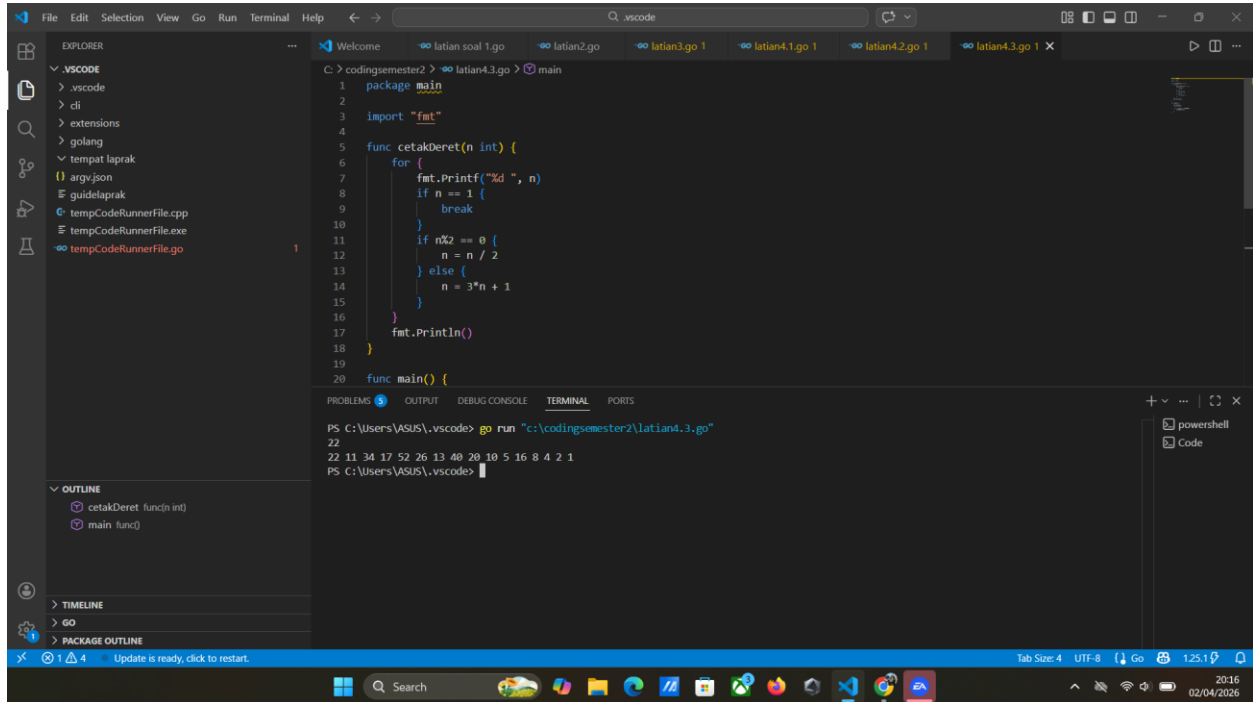
        if n%2 == 0 {
            n = n / 2
        } else {
            n = 3*n + 1
        }
    }

    fmt.Println()
}

func main() {
    var n int
    fmt.Scan(&n)

    if n > 0 && n < 1000000 {
        cetakDeret(n)
    }
}
```

Screenshot Code



The screenshot displays the Visual Studio Code (VS Code) interface with a Go program open in the editor. The program is a recursive function named `cetakDeret` that prints a sequence of numbers. The terminal shows the execution of the program, which outputs the sequence: 22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1.

```
1 package main
2
3 import "fmt"
4
5 func cetakDeret(n int) {
6     for {
7         fmt.Printf("%d ", n)
8         if n == 1 {
9             break
10        }
11        if n%2 == 0 {
12            n = n / 2
13        } else {
14            n = 3*n + 1
15        }
16    }
17    fmt.Println()
18 }
19
20 func main() {
```

Terminal Output:

```
PS C:\Users\ASUS\.vscode> go run "c:\codingsemester2\latian4.3.go"
22
22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1
PS C:\Users\ASUS\.vscode>
```

The interface also shows the Explorer sidebar with the file `tempCodeRunnerFile.go` selected, and the Outline sidebar showing the function `cetakDeret` and `main`.